IBM Developer
SKILLS NETWORK

Github link

Winning Space Race with Data Science

Data-Driven Insights to Improve First-Stage Reuse and Reduce Launch Costs

Context
Capstone Project – IBM Data Science

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Prepared by
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Date
Jan-April 2026

Github link

https://github.com/joriskamphuis/IBM-Data-Science-Final-Capstone-Project-2026/tree/main

Instructions and guidelines (read carefully)

Instructions

1. Insert your name as it appears on your Online Campus learner profile in the file name. Save the file as: **First Name Last Name M10U4 Capstone Project – e.g., Lilly Smith M10U4 Capstone Project.**
2. Write all your answers and upload your screenshots in this PowerPoint presentation in the slides provided.
3. Convert your final presentation into **.pdf** format and submit this **PDF document**. No other file types will be accepted.

Note: Please ensure that you have checked your course calendar for the due date for this assignment.

Guidelines

Note: If you do not have a desktop version, follow the instructions from the previous lab to access PowerPoint online.

1. Complete the slides with relevant details (you are free to add additional slides, charts, and tables).
2. Add any additional unused charts or tables to the appendix that you think are relevant to your analysis.

Plagiarism declaration
1. I understand that plagiarism is to use another's work and pretend that it is one's own. 2. I have not allowed, and will not allow, anyone to copy my work with the intention of submitting it as their own. 3. I acknowledge that copying someone else's assignment (or part of it) is wrong and declare that my assignments are my own work. **Plagiarism cases will be investigated in line with the Terms and Conditions for learners.**

Outline

Executive summary

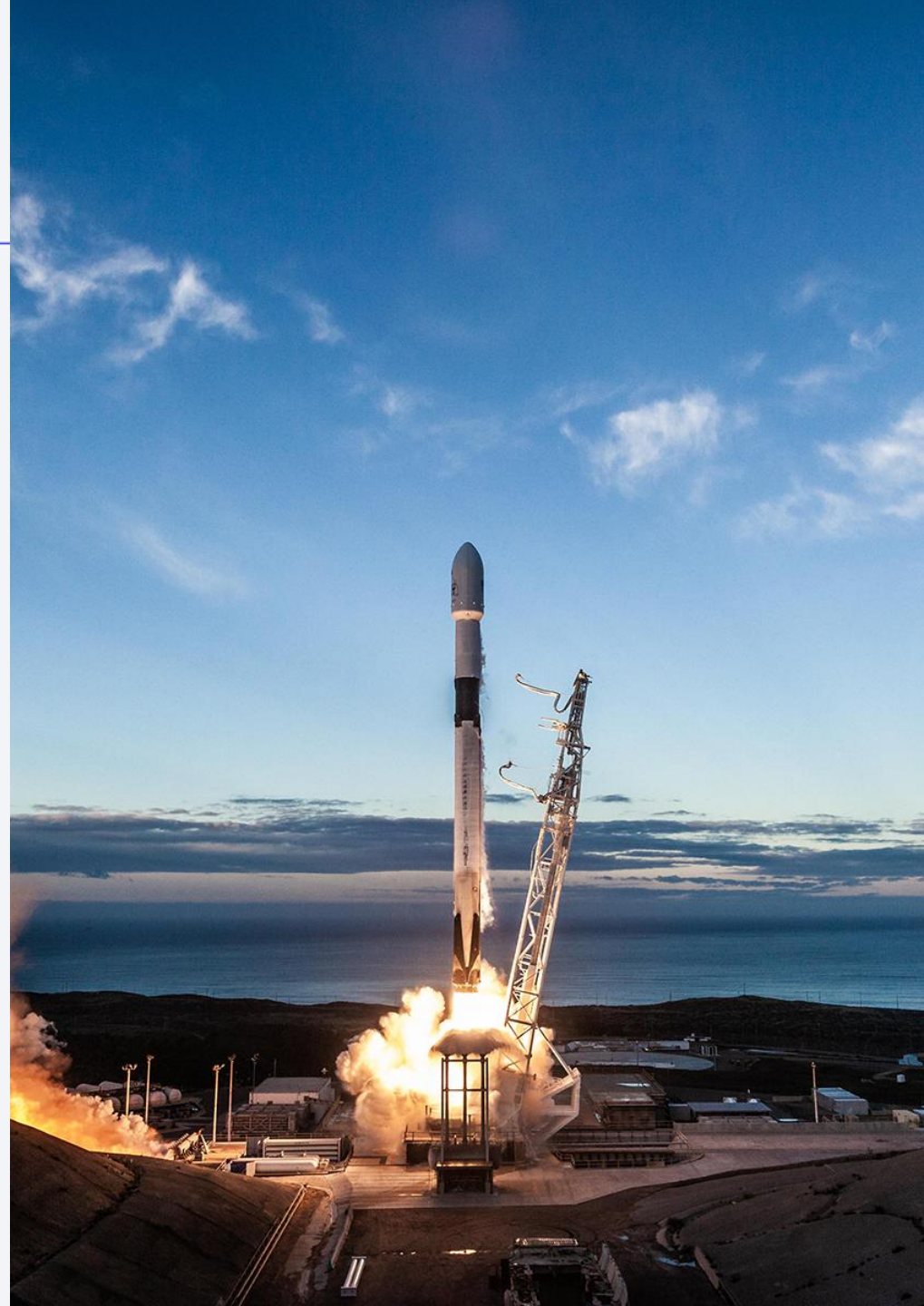
Introduction

Methodology

Results

Conclusion

Appendix



Executive summary (text)

This data science analysis applies the CRISP-DM methodology to examine historical rocket launch and landing data with the objective of supporting SpaceY's strategic ambition to compete with SpaceX through increased first-stage reusability, a key driver for reducing launch costs.

The project combines data collection, cleaning, exploratory analysis, geospatial visualization, and predictive modelling to better understand which factors are associated with successful rocket reuse. Interactive Folium maps and exploratory visualizations reveal clear operational patterns across launch sites, landing zones, orbit types, and mission profiles.

Several classification models were trained and evaluated to predict reuse success. However, the total number of historical launches available for training and testing is relatively limited, which restricts the statistical confidence with which one model can be identified as clearly superior. In addition, the analysis shows that reuse success increases over time, indicating that operational learning effects are present and that any static model is likely to experience model drift as SpaceY's capabilities evolve.

Despite these limitations, the modelling exercise provides valuable and consistent insights. Across four different predictive models, three produced highly similar conclusions (achieved an accuracy of 83%), suggesting that the identified relationships are robust rather than model-specific. In particular, the logistic regression model highlights rocket legs and grid fins as having the strongest absolute coefficient values, indicating that these recovery systems are critical enablers of successful reuse.

The primary value of this work lies not in predicting individual launch outcomes with high certainty, but in guiding strategic and engineering focus. The results strongly suggest that investments in landing hardware, recovery control, and reuse-enabling design features are essential for improving reuse rates and lowering cost per launch, thereby strengthening SpaceY's long-term competitive position.

Classification: PUBLIC

Executive summary (bullets)

Objective: Support SpaceY's competitiveness through improved rocket reusability and lower launch costs

Approach: Applied CRISP-DM using launch data, EDA, Folium maps, and predictive models

Insights: Reuse success depends on combined mission, launch, and landing factors

Modeling: Limited sample size; increasing success over time implies model drift risk

Performance: 3 of 4 models achieved ~83% accuracy, showing consistent predictive behaviour

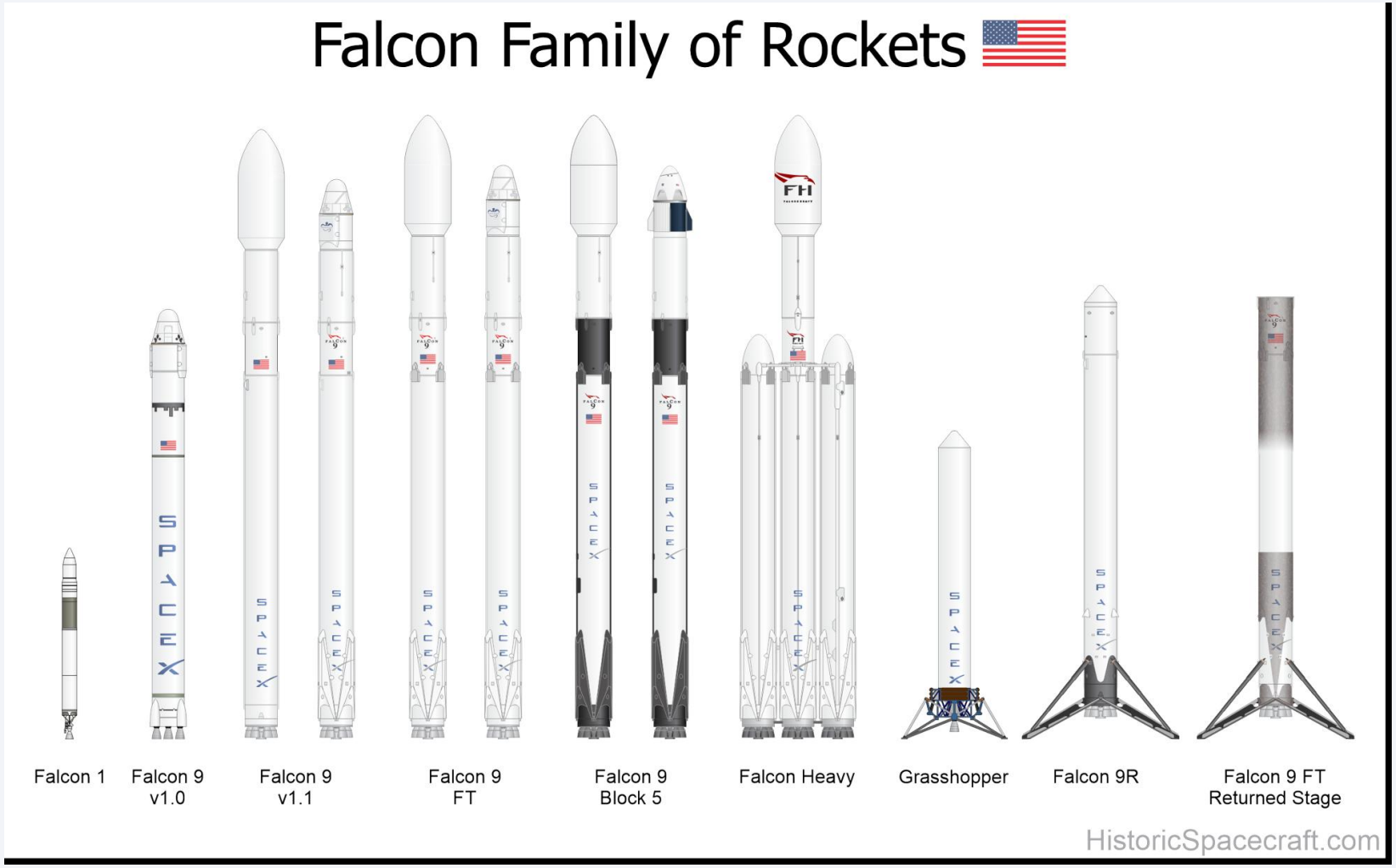
Key Driver: Rocket legs & grid fins show strongest impact on reuse success

Conclusion: Models inform where to focus engineering and operations, not absolute prediction

Introduction

Background
Reusing rockets drops cost of launch

Problem you want to solve
Predict if SpaceX Falcon 9 rocket's first stage will land successfully



Classification: PUBLIC

Introduction

Recovered SpaceX Falcon 9 Booster Moves Back to KSC for Eventual Reflight

Classification: PUBLIC

Section 1

Methodology

Classification: PUBLIC

Methodology

Executive summary

- Data collection methodology
  - Describe how the data was collected.
- Perform data wrangling.
  - Describe how the data was processed.
- Perform exploratory data analysis (EDA) using visualization and SQL.
- Perform interactive visual analytics using Folium and Plotly Dash.
- Perform predictive analysis using classification models.
  - How to build, tune, and evaluate classification models

Classification: PUBLIC

Data collection

Data Sources

- Primary source: SpaceX REST API (v4)
- Time horizon: Launches up to 13 November 2020

Tools & Technologies

- Python
- SpaceX REST API (HTTP GET)
- Pandas, NumPy
- JSON data format

Data Observations

- Payload mass contained missing values → Replaced with mean payload mass
- Landing pad legitimately missing → Kept as missing (represents no landing attempt)

Flowchart (High Level)

Start
↓
Request SpaceX launch data (API)
↓
Parse JSON response
↓
Normalize into tabular format
↓
Filter relevant launches
↓
Enrich data using API lookups
↓
Handle missing values
↓
Create final analysis data set
↓
End

Classification: PUBLIC

Data collection – SpaceX API

- The dataset was collected using RESTful HTTP GET requests to the SpaceX API, where launch records were retrieved first and then enriched through additional REST calls using resource IDs.

Notebook URL:

https://github.com/joriskamphuis/IBM-Data-Science-Final-Capstone-Project-2026/blob/main/jupyter-labs-spacex-data-collection-api%20(1).ipynb

Flowchart of SpaceX API calls

Start
↓
HTTP GET/launches/past
↓
Launch Data (JSON)
↓
Extract IDs(rocket, payload, launchpad, core)
↓
HTTP GET Requests
/rockets/{id}
/payloads/{id}
/launchpads/{id}
/cores/{id}
↓
JSON Responses
↓
Parse & Append Data
↓
Create Enriched Dataset
↓
End

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Data collection – Scraping (from Wikipedia)

- Web scraping using BeautifulSoup
- HTTP GET request to Wikipedia (static page)
- HTML table extraction
- Parsing and cleaning table cells
- Structured data storage in Pandas
- CSV export for downstream analysis

Notebook URL:

https://github.com/joriskamphuis/IBM-Data-Science-Final-Capstone-Project-2026/blob/main/jupyter-labs-webscraping%20(1).ipynb

Data table URL:

https://github.com/joriskamphuis/IBM-Data-Science-Final-Capstone-Project-2026/blob/main/spacex_web_scraped%20(1).csv

Flowchart of Web Scraping

Start
↓
HTTP GET Request (Wikipedia page)
↓
HTML Page Content
↓
Parse HTML with BeautifulSoup
↓
Locate Falcon 9 Launch Table
↓
Extract Rows & Cells
↓
Clean & Parse Data (Date, Booster, Payload, Orbit, Outcome)
↓
Create Pandas DataFrame
↓
Export to CSV End

Classification: PUBLIC

Data collection – Scraping (from Wikipedia)

IBM-Data-Science-Final-Capstone-Project-2026 / spacex_web_scraped (1).csv

joriskamphuis 9.4.4 Data Collection with Web Scraping lab 1344751 · 2 days ago History

Preview Code Blame 214 lines (214 loc) · 11.5 KB Raw

Search this file

Flight No.	Launch site	Payload	Payload mass	Orbit	Customer	Launch outcome	Version Booster	Booster landing	Date	Time
1	CCAFS	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	F9 v1.07B0003.18	Failure	4 June 2010	18:45
2	CCAFS	Dragon	0	LEO	NASA	Success	F9 v1.07B0004.18	Failure	8 December 2010	15:43
3	CCAFS	Dragon	525 kg	LEO	NASA	Success	F9 v1.07B0005.18	No attempt	22 May 2012	07:44
4	CCAFS	SpaceX CRS-1	4,700 kg	LEO	NASA	Success	F9 v1.07B0006.18	No attempt	8 October 2012	00:35
5	CCAFS	SpaceX CRS-2	4,877 kg	LEO	NASA	Success	F9 v1.07B0007.18	No attempt	1 March 2013	15:10
6	VAFB	CASSIOPE	500 kg	Polar orbit	MDA	Success	F9 v1.17B10038	Uncontrolled	29 September 2013	16:00
7	CCAFS	SES-8	3,170 kg	GTO	SES	Success	F9 v1.1	No attempt	3 December 2013	22:41
8	CCAFS	Thaicom 6	3,325 kg	GTO	Thaicom	Success	F9 v1.1	No attempt	6 January 2014	22:06
9	Cape Canaveral	SpaceX CRS-3	2,296 kg	LEO	NASA	Success	F9 v1.1	Controlled	18 April 2014	19:25
10	Cape Canaveral	Orbcomm-OG2	1,316 kg	LEO	Orbcomm	Success	F9 v1.1	Controlled	14 July 2014	15:15
11	Cape Canaveral	AsiaSat 8	4,535 kg	GTO	AsiaSat	Success	F9 v1.1	No attempt	5 August 2014	08:00
12	Cape Canaveral	AsiaSat 6	4,428 kg	GTO	AsiaSat	Success	F9 v1.1[	No attempt	7 September 2014	05:00
13	Cape Canaveral	SpaceX CRS-4	2,216 kg	LEO	NASA	Success	F9 v1.1[	Uncontrolled	21 September 2014	05:52
14	Cape Canaveral	SpaceX CRS-5	2,395 kg	LEO	NASA	Success	F9 v1.1[	Failure	10 January 2015	09:47
15	Cape Canaveral	DSCOVR	570 kg	HEO	USAF	Success	F9 v1.1[	Controlled	11 February 2015	23:03
16	Cape Canaveral	ABS-3A	4,159 kg	GTO	ABS	Success	F9 v1.1[	No attempt	2 March 2015	03:50

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Section 2

Insights drawn from EDA

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EDA with SQL

QL Tasks Performed

- Loaded SpaceX launch dataset into a **SQLite / Db2 database**
- Created structured table for querying (**SPACEXTABLE**)
- Queried distinct **launch sites**
- Filtered launches by **launch site and date**
- Calculated **total and average payload mass**
- Identified **successful landing dates and conditions**
- Analyzed **mission success vs failure counts**
- Retrieved launches with **maximum payload mass**
- Filtered launches by **year and landing outcome**
- Ranked **landing outcomes by frequency** for different time periods

GitHub URL: https://github.com/joriskamphuis/IBM-Data-Science-Final-Capstone-Project-2026/blob/main/jupyter-labs-eda-sql-edx_sqllite%20(1).ipynb

Unique Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

BETWEEN '2010-06-04' **AND** '2017-03-20'

Landing_Outcome	Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

BETWEEN '2017-03-20' **AND** '2020/12/12'

Landing_Outcome	Count
Success	38
No attempt	11
Success (drone ship)	9
Success (ground pad)	6
Failure	3
Controlled (ocean)	2
No attempt	1

Classification: PUBLIC

Section 3

Launch Sites Proximities Analysis

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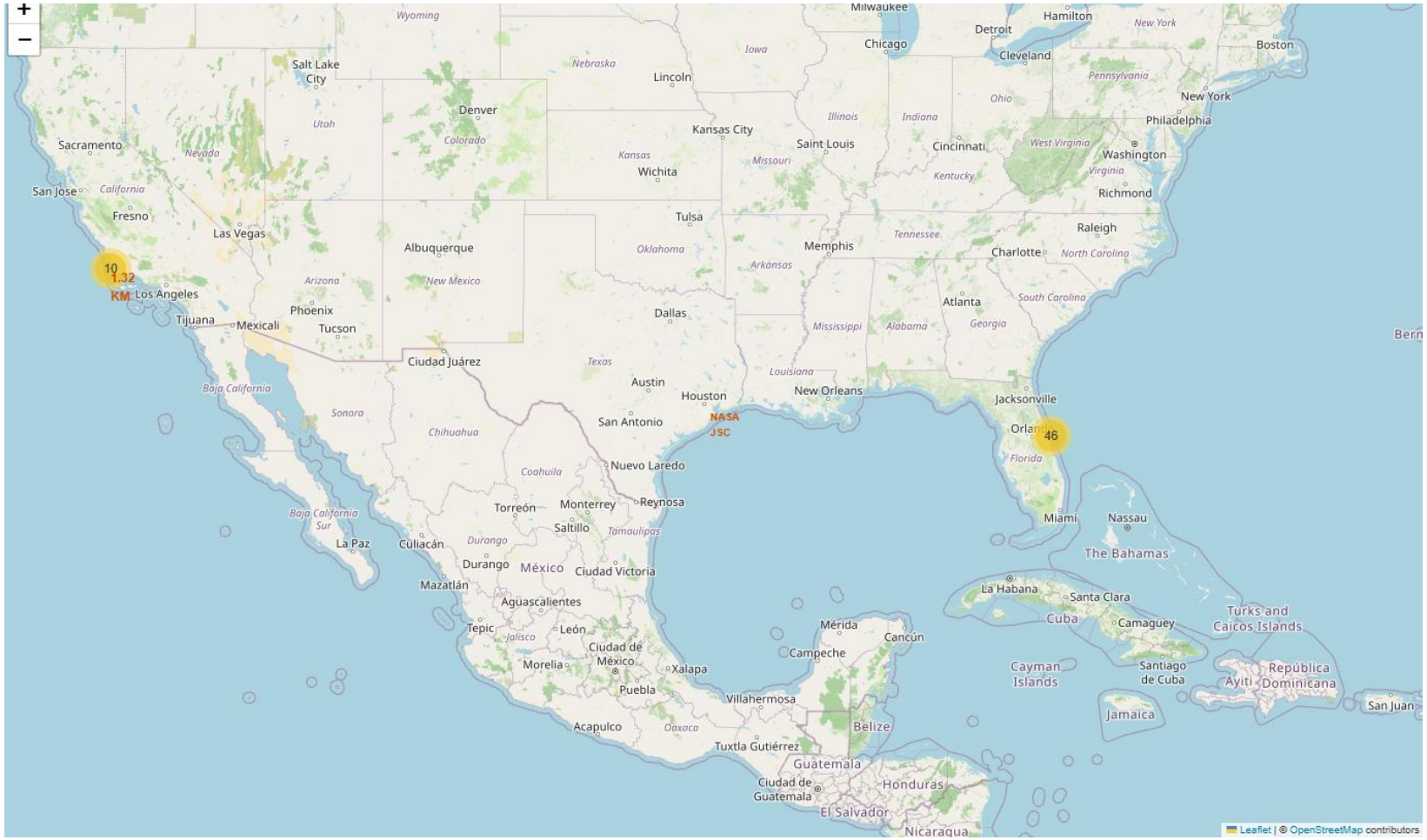
Launch site's location + number of launches



Interactive map with Folium

GitHub URL: https://github.com/joriskamphuis/IBM-Data-Science-Final-Capstone-Project-2026/blob/main/lab_jupyter_launch_site_location%20(4).ipynb

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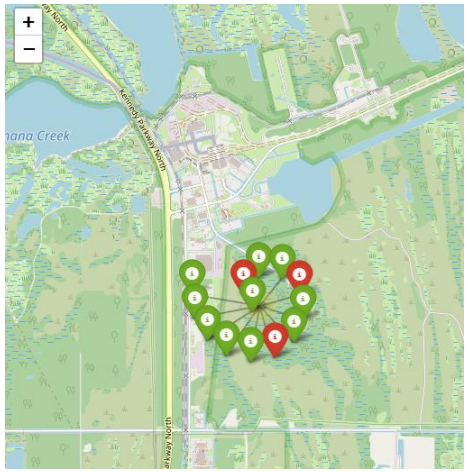
of successful launches per location

Biased comparison?
10 launches on the West Coast
vs 46 on the East Coast



Launch site in a forest?

GitHub URL: https://github.com/joriskamphuis/IBM-Data-Science-Final-Capstone-Project-2026/blob/main/lab_jupyter_launch_site_location%20(4).ipynb

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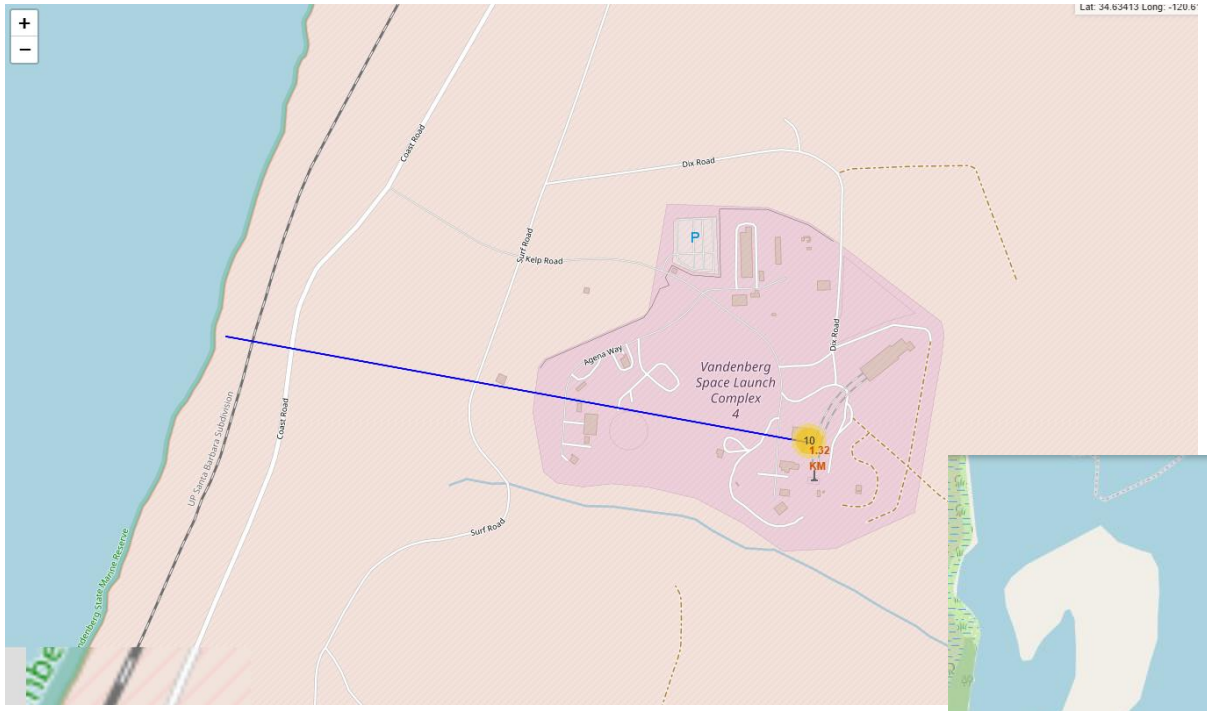
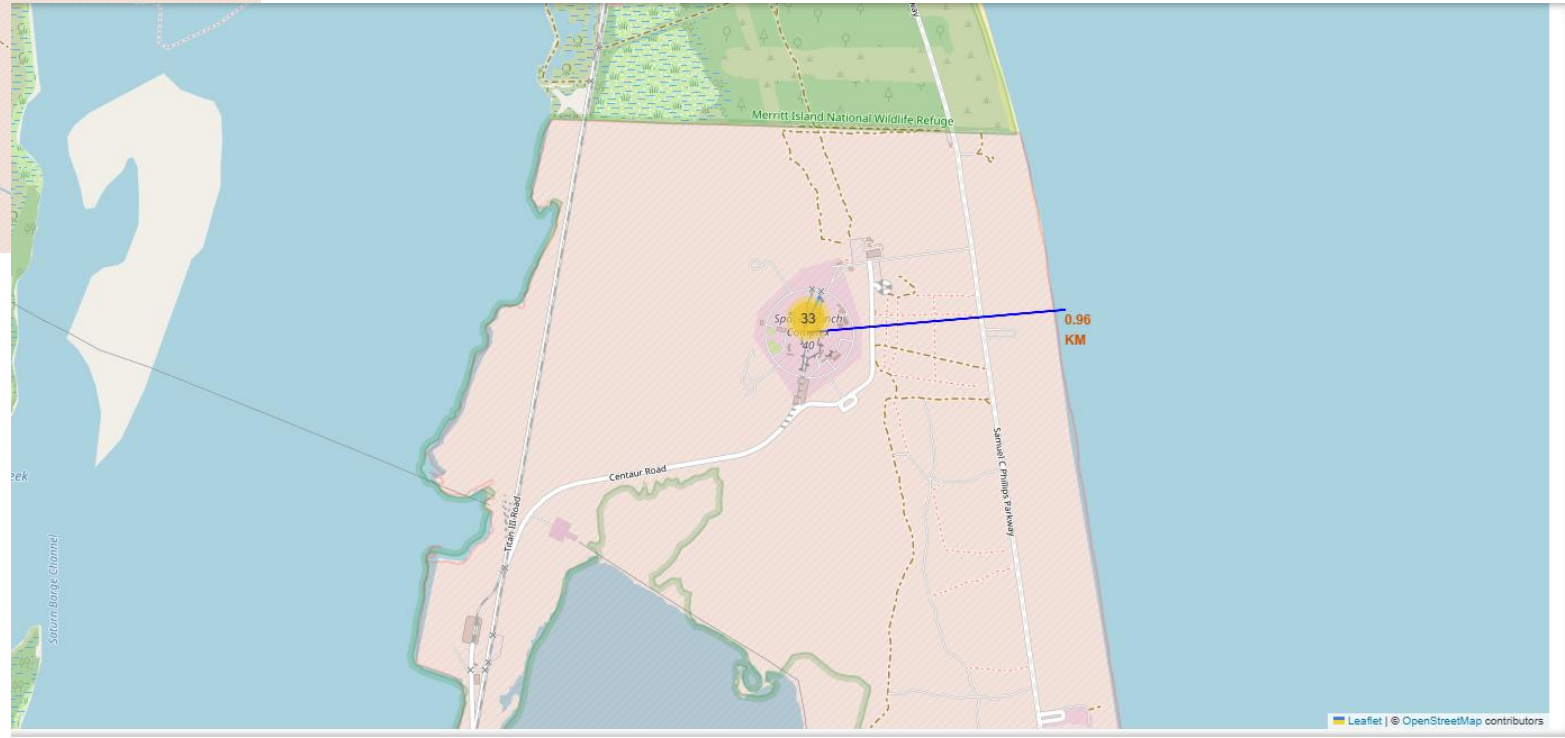
Distance to sea per launch location



1 km for the two launch sites

Reasons (suggestions):

- Safety (no civilisation close by)
- Logistics and transportation
- Cooling water nearby



GitHub URL: https://github.com/joriskamphuis/IBM-Data-Science-Final-Capstone-Project-2026/blob/main/lab_jupyter_launch_site_location%20(4).ipynb

Section 4

Build a Dashboard with Plotly Dash

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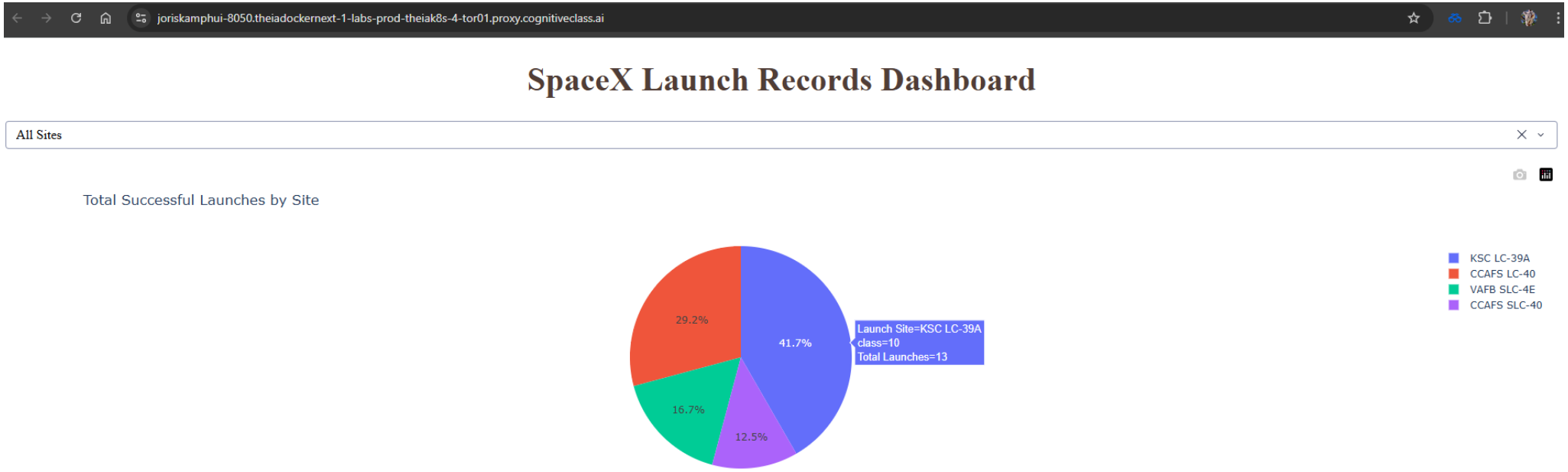
Build a dashboard with Plotly Dash

- Summarize the plots and interactions you added to a dashboard.
- Explain why you added those plots and interactions.
- Add the GitHub URL of your completed Plotly Dash lab as an external reference.

Classification: PUBLIC

Build a dashboard with Plotly Dash

Displays all successful launches grouped by sites

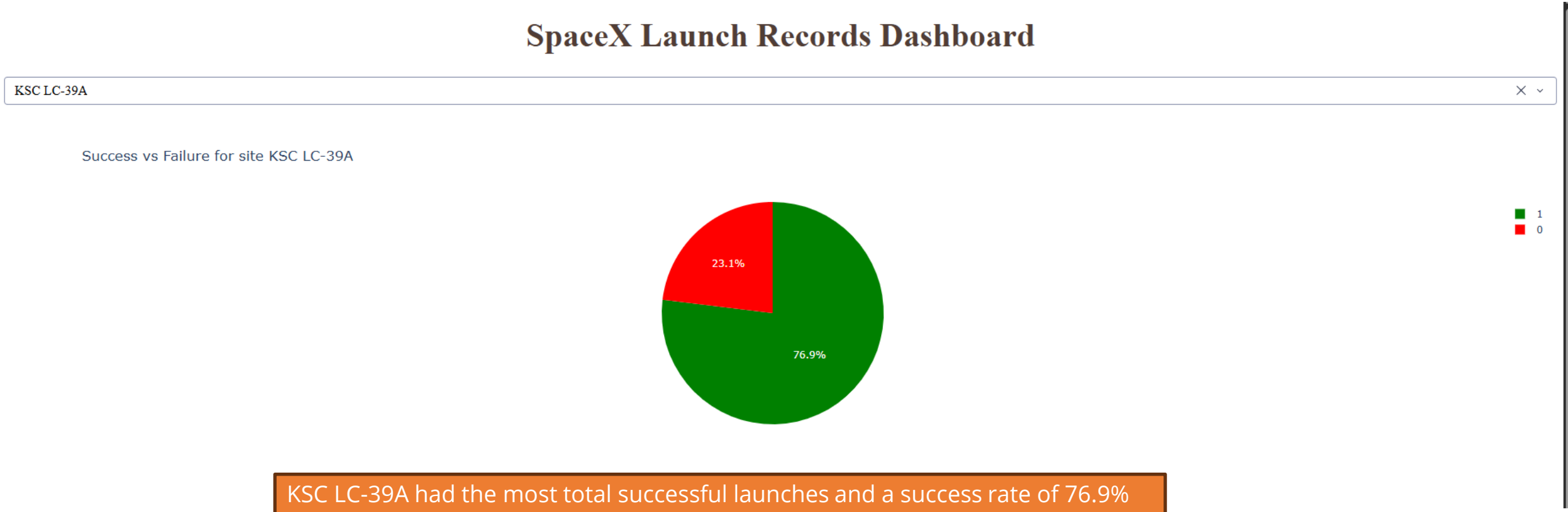


GitHub URL:
https://github.com/joriskamphuis/IBM-Data-Science-Final-Capstone-Project-2026/blob/main/spacex-dash-app.py

Classification: PUBLIC

Build a dashboard with Plotly Dash

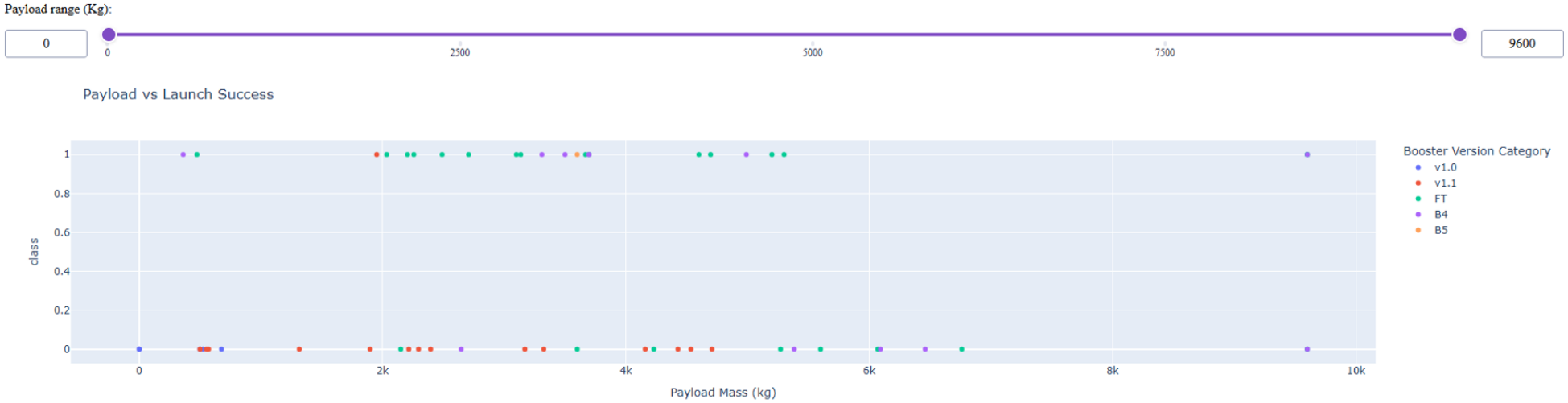
Displays per site percentages of successful and failed launches



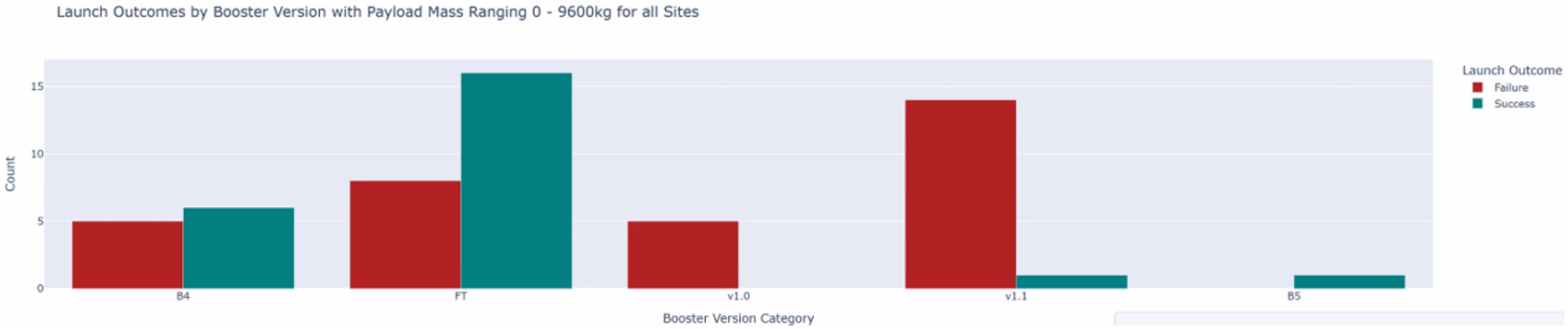
KSC LC-39A had the most total successful launches and a success rate of 76.9%
Other launch site may have higher or lower success rate, but samples size is smaller

Classification: PUBLIC

Build a dashboard with Plotly Dash



No correlation between payload mass and success or unsuccessful launches



Karens' improved visualisation of influence of Booster version on success rate

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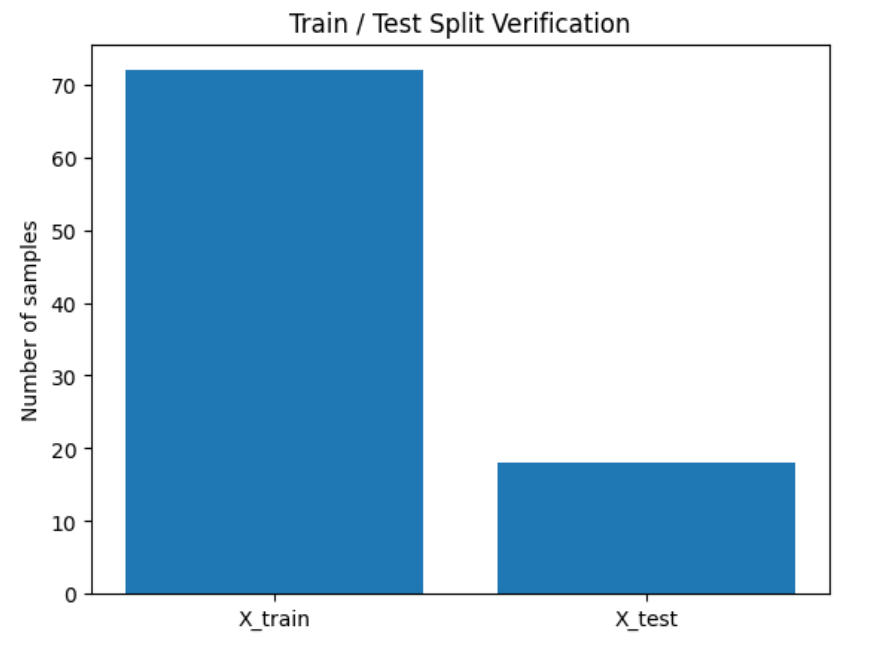
Section 5

Predictive Analysis (Classification)

Classification: PUBLIC

Predictive analysis (classification)

Problem Definition
Objective: Predict whether the Falcon 9 first stage will successfully land
Target variable:
Class (1 = Success, 0 = Failure)



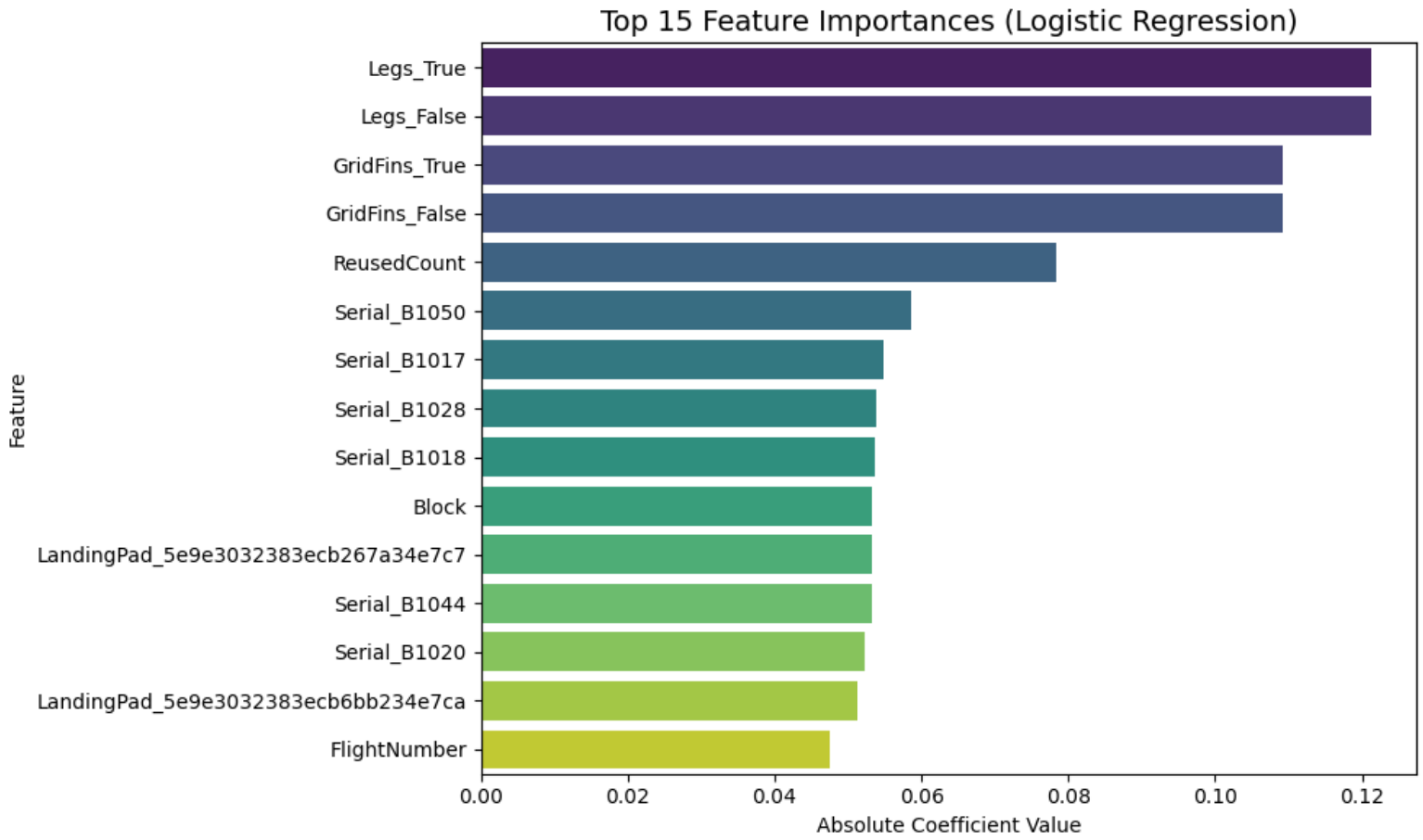
GitHub URL https://github.com/joriskamphuis/IBM-Data-Science-Final-Capstone-Project-2026/blob/main/SpaceX_Machine%20Learning%20Prediction_Part_5%20(3).ipynb

Start
↓
Prepared Dataset (Cleaned & Engineered Features)
↓
Feature Selection & Encoding (Numerical + One-Hot Encoding)
↓
Train / Test Split (Scaling/normalization done after before splitting!)
↓
Model Training (**Logistic Regression, SVM, Decision Tree, KNN**)
↓
Hyperparameter Tuning (GridSearchCV + Cross-Validation)
↓
Model Evaluation (Test Accuracy, Confusion Matrix)
↓
Model Comparison
↓
Best Model Selected

End

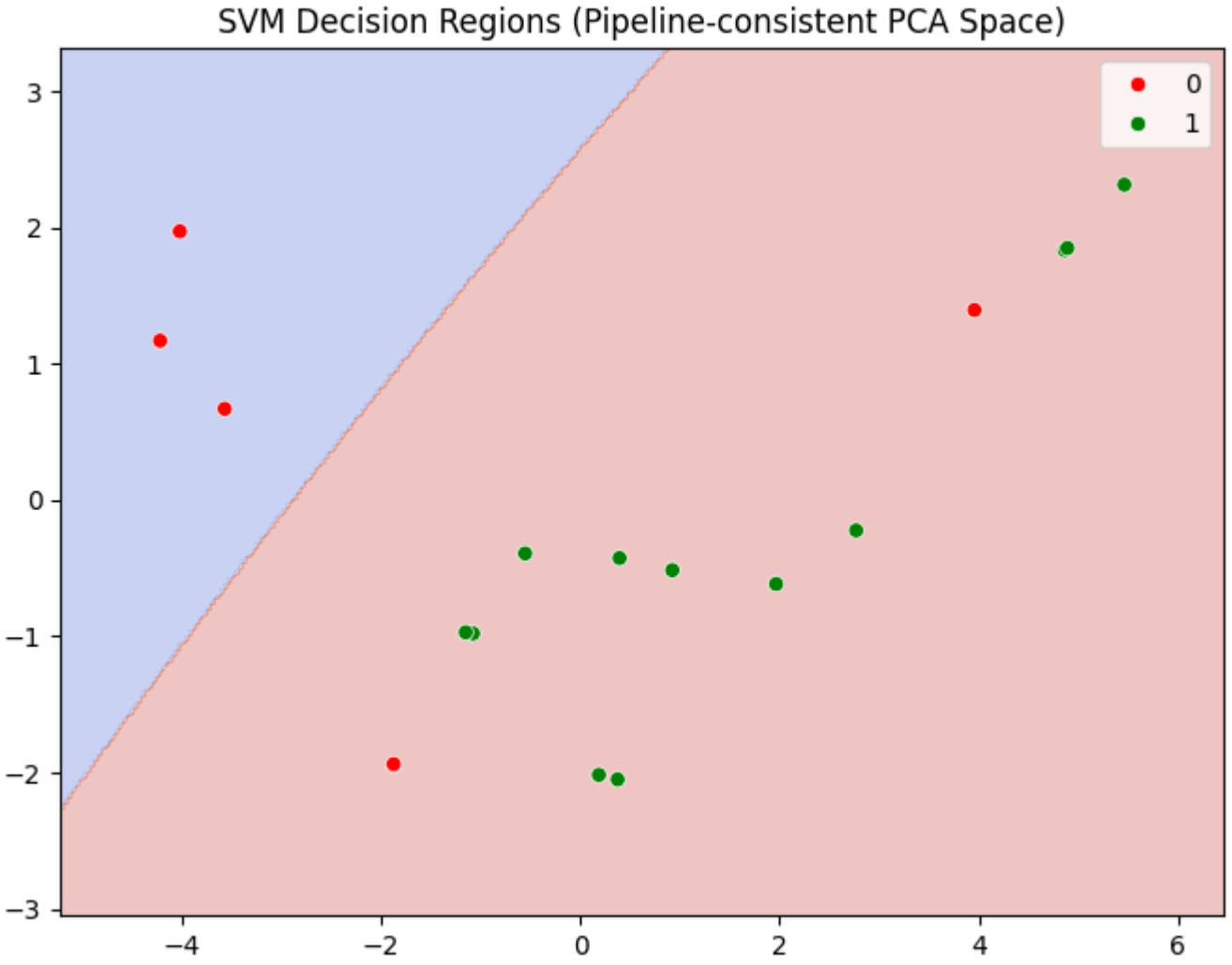
Classification: PUBLIC

Predictive analysis (classification)
What features influence the success rate?



Classification: PUBLIC

Predictive analysis (classification)
How is the SVM model deciding?



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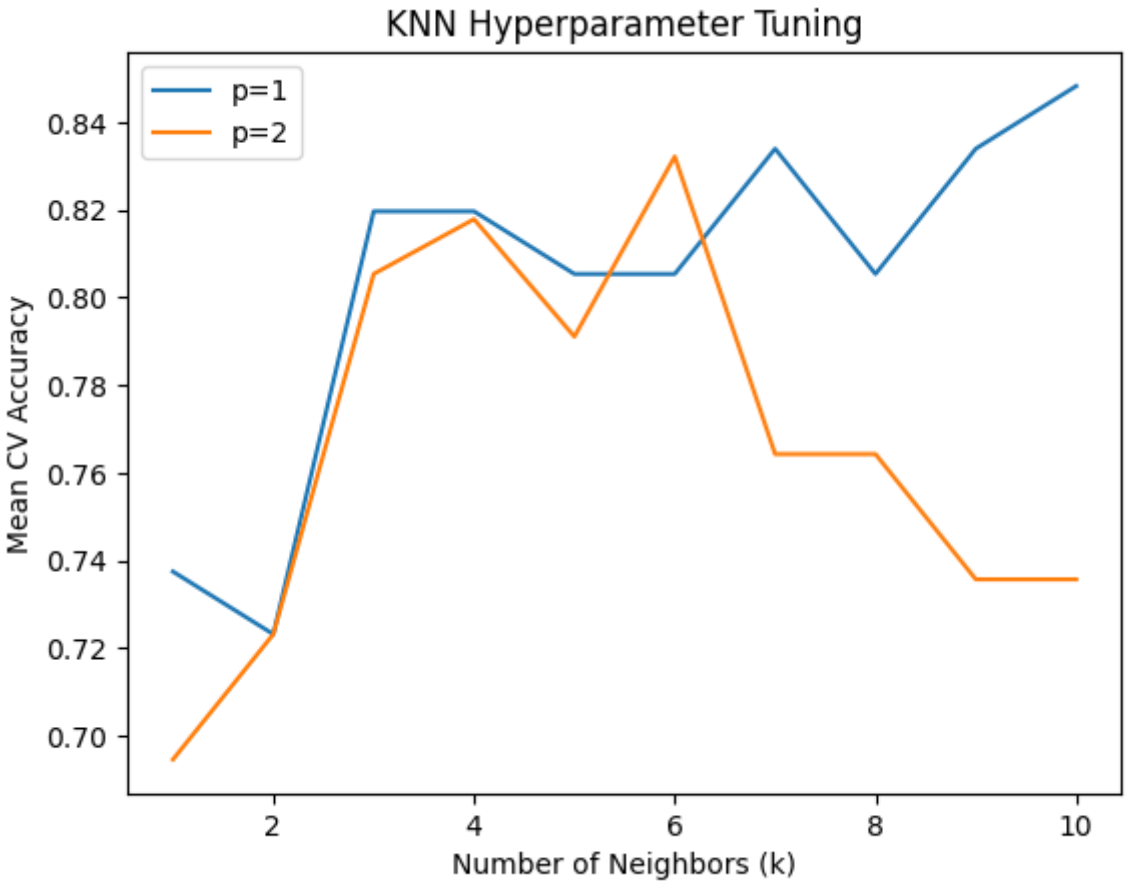
Predictive analysis (classification)
How is the KNN model tuned?



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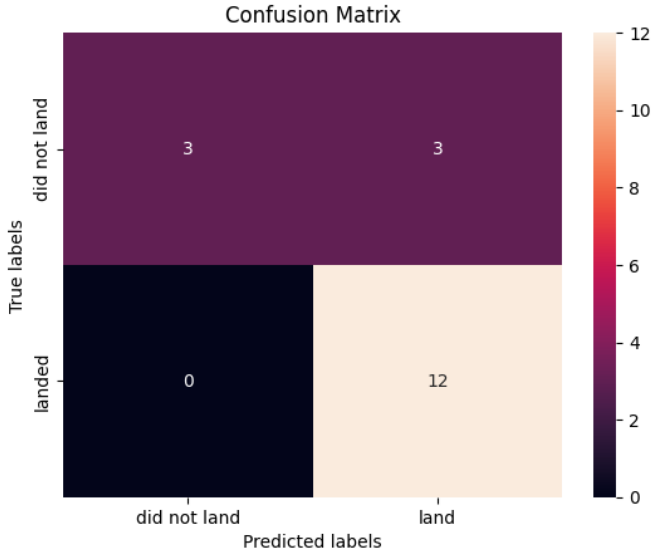
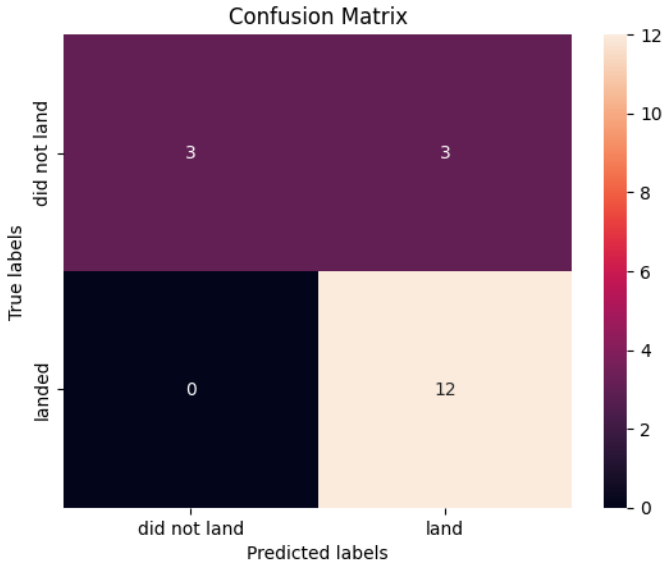
Predictive analysis (classification) Performance Analysis

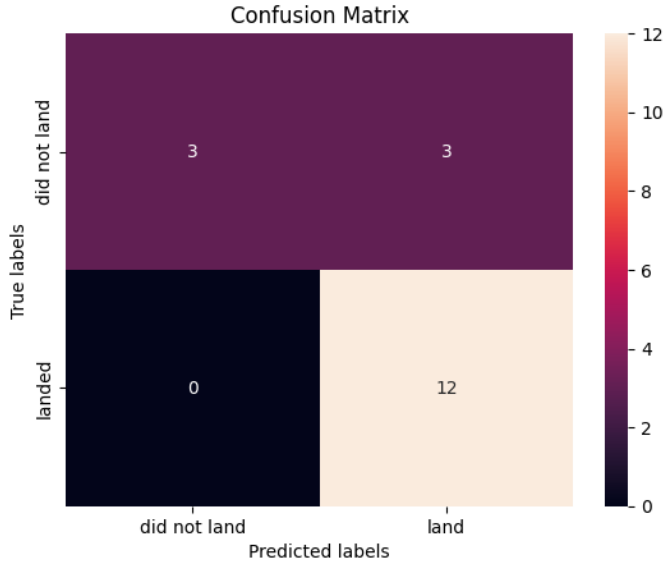
```
yhat=logreg_cv.predict(X_test)
plot_confusion_matrix(Y_test,yhat)
```



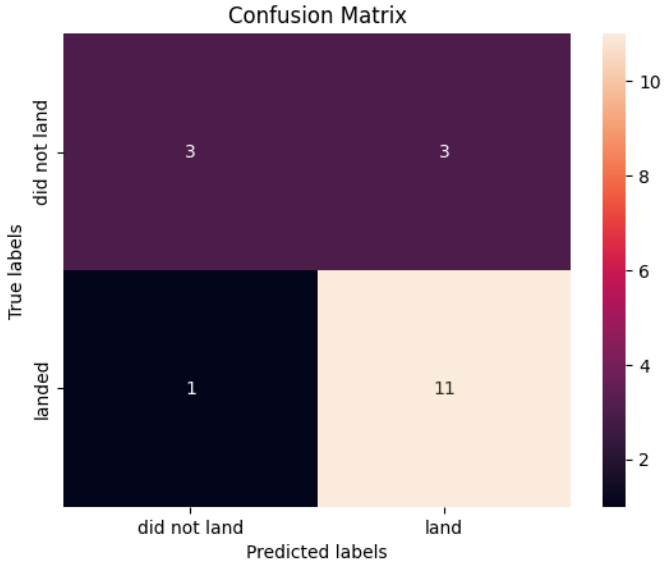

```
yhat=svm_cv.predict(X_test)
plot_confusion_matrix(Y_test,yhat)
```




```
yhat = knn_cv.predict(X_test)
plot_confusion_matrix(Y_test,yhat)
```




```
yhat = tree_cv.predict(X_test)
plot_confusion_matrix(Y_test,yhat)
```



3 models with similar confusion matrices:

- Small sample size
- Small differences in performance
- No difference in confusion matrix

Further investigate the 3 false negative launches

Only Decision Tree model performed worse with one extra false positive

Classification: PUBLIC

Predictive analysis (classification) Performance Analysis

Logistic Regression

ROC AUC score: 0.888888888888889

	precision	recall	f1-score	support
0	1.00	0.50	0.67	6
1	0.80	1.00	0.89	12
accuracy			0.83	18
macro avg	0.90	0.75	0.78	18
weighted avg	0.87	0.83	0.81	18

Support Vector Machine

ROC AUC score: 0.8472222222222223

	precision	recall	f1-score	support
0	1.00	0.50	0.67	6
1	0.80	1.00	0.89	12
accuracy			0.83	18
macro avg	0.90	0.75	0.78	18
weighted avg	0.87	0.83	0.81	18

Decision Tree

ROC AUC score: 0.8472222222222222

	precision	recall	f1-score	support
0	0.75	0.50	0.60	6
1	0.79	0.92	0.85	12
accuracy			0.78	18
macro avg	0.77	0.71	0.72	18
weighted avg	0.77	0.78	0.76	18

K-Nearest Neighbor

ROC AUC score: 0.8472222222222222

	precision	recall	f1-score	support
0	1.00	0.50	0.67	6
1	0.80	1.00	0.89	12
accuracy			0.83	18
macro avg	0.90	0.75	0.78	18
weighted avg	0.87	0.83	0.81	18

Logistic Regression achieves the highest ROC accuracy score and is therefore selected as the best* overall model.
*performance differences between models are small and results should be interpreted with caution

Classification: PUBLIC

Conclusions

- Successfully collected and integrated SpaceX data from multiple sources
- Identified key factors influencing Falcon 9 landing success
- Prioritize investment in recovery hardware, particularly rocket legs and grid fins, as key enablers of reuse
- Built and evaluated multiple predictive models
- Achieved ~83% prediction accuracy with strong discriminatory power
- Results demonstrate feasibility of predicting launch outcomes

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Recommondations Work

- Implement periodic model retraining (data drift mitigation)
- Expand feature data set to improve models' accuracy
- Model a Recurrent Neural Network (RNN) with time sequence modeling

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Recommondations: Reconnect Analysis to the Business Goal

- Do not evaluate the model in isolation or focus solely on prediction accuracy
- The core business objective is to compete with SpaceX by increasing rocket reusability
- Higher reusability enables lower launch costs and long-term competitiveness for SpaceY
- Model insights should only be assessed based on how well they support decisions that improve reusability and cost reduction

Classification: PUBLIC

Appendix

- All codes used and data tables can be found in the GitHub repository:
- https://github.com/joriskamphuis/IBM-Data-Science-Final-Capstone-Project-2026/tree/main
- Microsoft Copilot (GPT-5 based) was used as a support tool for coding assistance and text refinement (Final decisions, conclusions, and recommendations are author-owned)

Classification: PUBLIC

Thank you!

Classification: PUBLIC